KIRUBIA A

Data Analyst | Data Scientist | ML Engineer | AI & Data Science

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PROFESSIONAL SUMMARY

Results-driven Data Analyst and ML Engineer with B.Tech in Artificial Intelligence & Data Science (CGPA 8.2). Hands-on experience building end-to-end machine learning pipelines, performing exploratory data analysis (EDA), and deploying interactive dashboards. Proficient in Python, SQL, Pandas, Scikit-learn, Power BI, and Tableau. Demonstrated ability to extract actionable insights from large datasets, build predictive models, and communicate findings to stakeholders. Seeking a data analyst, data scientist, or ML engineering role to drive data-driven decision-making at scale.

TECHNICAL SKILLS

Languages & Data: Python (Pandas, NumPy, SciPy), SQL (MySQL — joins, window functions, subqueries, CTEs), R basics, Excel (Pivot Tables, Power Query, XLOOKUP)

Machine Learning: Scikit-learn, XGBoost, LightGBM, Random Forest, Logistic Regression, KNN, SVM, K-Means, DBSCAN, PCA, SHAP, SMOTE, GridSearchCV, Cross-Validation

NLP & AI: TF-IDF, VADER Sentiment Analysis, NLTK, spaCy, Hugging Face Transformers, LangChain basics, Prompt Engineering, RAG pipeline basics

BI & Visualisation: Power BI (DAX, KPI cards, slicers, interactive dashboards), Tableau, Matplotlib, Seaborn, Plotly, Streamlit

MLOps & Cloud: MLflow, FastAPI, Docker basics, Git, GitHub, GCP (BigQuery, Colab), CI/CD basics, Jupyter Notebook, VS Code, Postman

Core Concepts: EDA, Feature Engineering, Hypothesis Testing, A/B Testing, ETL, Data Cleaning, Statistical Analysis, Data Storytelling, Stakeholder Communication

PROFESSIONAL EXPERIENCE

Data Science Intern | *Gradtwin Services* | Sept 2024 – Nov 2024

- Performed **exploratory data analysis (EDA)** on real-world datasets — identified trends, patterns, and business-relevant outliers using Pandas, Matplotlib, and Seaborn
- Built and benchmarked supervised ML models (**KNN, Random Forest**) using **cross-validation and GridSearchCV**; improved model accuracy by 12% through hyperparameter tuning
- Engineered features via label encoding, one-hot encoding, normalisation, and transformation pipelines; evaluated performance using **F1-score, ROC-AUC, precision, recall, and confusion matrix**
- Translated findings into stakeholder-ready reports with clear visualisations and data-driven recommendations

PROJECTS

Nykaa Beauty Brand Review Analysis *Python · Pandas · SQL · Matplotlib · Seaborn · VADER · Power BI · Git · GitHub*

- Analysed **61,284 customer reviews** across 11 top beauty brands — end-to-end pipeline: data cleaning, feature engineering, EDA, SQL querying, visualisation, and interactive dashboard
- Applied **VADER sentiment analysis** on 61k review texts; identified Kay Beauty as top-rated brand (4.69 avg rating, highest 5-star ratio); review volume tripled in 2019 aligned with pre-IPO growth
- Built **9-visual interactive Power BI dashboard** with brand slicers, KPI cards, trend charts; insight: medium discounts (11–25%) drive higher review engagement than high discounts — published on GitHub

Global AI Adoption & Workforce Impact *XGBoost · LightGBM · K-Means · SHAP · MLflow · Streamlit · SMOTE · Python*

- Engineered full **ML pipeline on 150,000+ records** — XGBoost multi-class classifier with SHAP explainability + K-Means clustering for AI maturity segmentation across 20+ industries
- Tracked all experiments via **MLflow** (parameters, metrics, model artefacts); handled class imbalance with **SMOTE**; deployed interactive Streamlit dashboard with real-time filters

Sentiment Analysis — Product Reviews *Python · NLTK · spaCy · TF-IDF · Logistic Regression · Scikit-learn · NLP*

- Built end-to-end **NLP pipeline**: text cleaning, tokenisation, stopword removal, TF-IDF vectorisation → Logistic Regression classifier; evaluated with precision, recall, F1-score, and confusion matrix

Sales Data Analysis & KPI Dashboard | India Housing Price Analysis *Python · SQL · Tableau · Excel · Pandas · Seaborn · Regression · Streamlit*

- Sales: Analysed **9,800+ records** with SQL and Python; built Tableau KPI dashboards; identified excessive discounting as net revenue drain — delivered corrective recommendations to stakeholders
- Housing: Conducted **correlation analysis, pair plots, statistical tests**; identified location and property type as strongest pricing predictors; deployed Streamlit dashboard with dynamic regional filters

Cirrhosis Disease Stage Prediction *Python · Scikit-learn · Random Forest · Logistic Regression · SMOTE · ROC-AUC*

- Built **multi-class classifier** on clinical patient data (Random Forest outperformed baseline by 18%); applied SMOTE for class imbalance; evaluated with accuracy, F1-score, and AUC-ROC across all stages

EDUCATION

B.Tech – Artificial Intelligence & Data Science | CGPA: 8.2 | 2021 – 2025

SIMATS School of Engineering, Chennai | Relevant Coursework: Machine Learning, Deep Learning, NLP, Statistics, Database Systems, Data Visualisation, Algorithms

CERTIFICATIONS

- Accenture Data Analytics & Visualisation Job Simulation | Forage | Aug 2024
- Oracle Cloud Infrastructure GenAI Certified Professional | Oracle | Jul 2024
- Oracle Cloud Infrastructure AI Foundations Associate | Oracle | Aug 2024